

Algorithm Comparison: Causal Discovery for Bias Detection

Algorithm	SHD Biased	SHD Unbiased	Race→Loan (Biased)	Race→Loan (Unbiased)	Handles Latent	Gives β
PC	5	2	No	OK ✓	No	No
FCI (full)	7	7	No	OK ✓	Yes	No
GES	3	2	No	OK ✓	No	No
GRaSP	3	2	No	OK ✓	No	No
ICA-LiNGAM	6	5	Yes ✓	FP ✗	No	Yes
DirectLiNGAM	6	4	Yes ✓	OK ✓	No	Yes